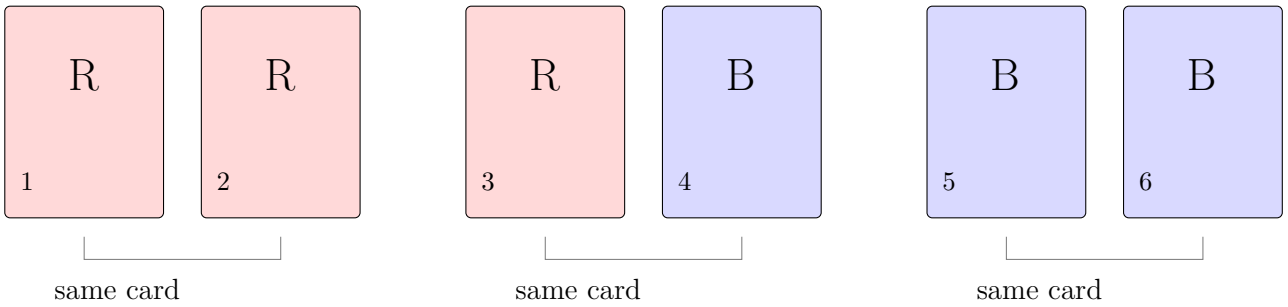
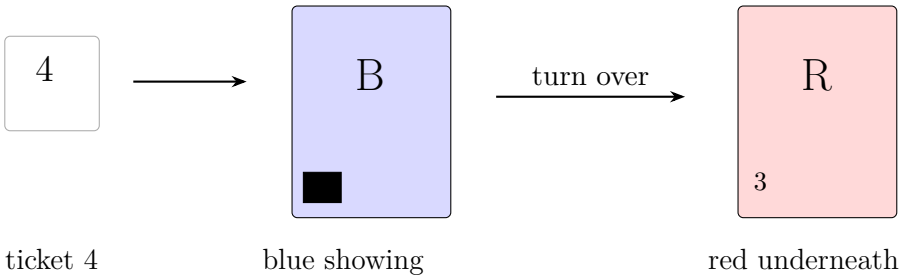
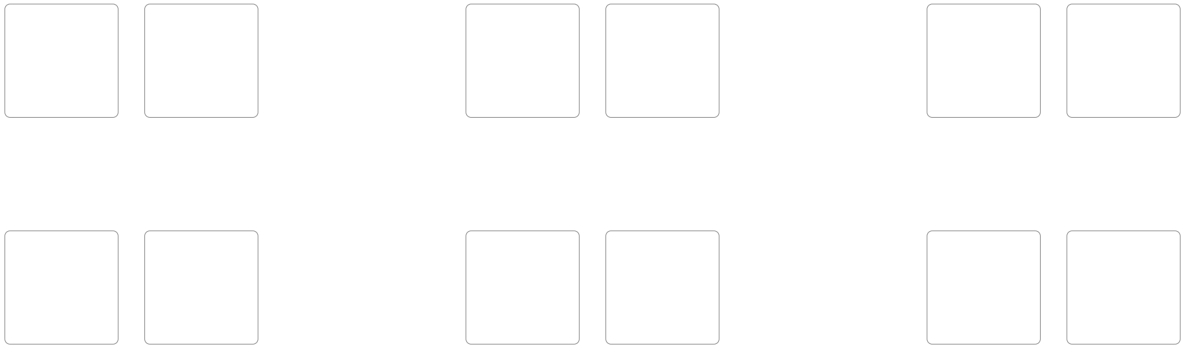
The small numbers match the six tickets. A secret ticket picks the face that shows, with its number covered. You see only the color. Every ticket has the same chance.





Problem 1: Find every pair of tickets that show the same color but hide different colors.



Problem 2: You are told only that red is showing. Which hidden color is the better guess, and why?

red underneath

blue underneath

Problem 3: You are told only that blue is showing. Does the better guess change?

red underneath

blue underneath

Only tickets 1 and 3 are in the cup, each with the same chance.



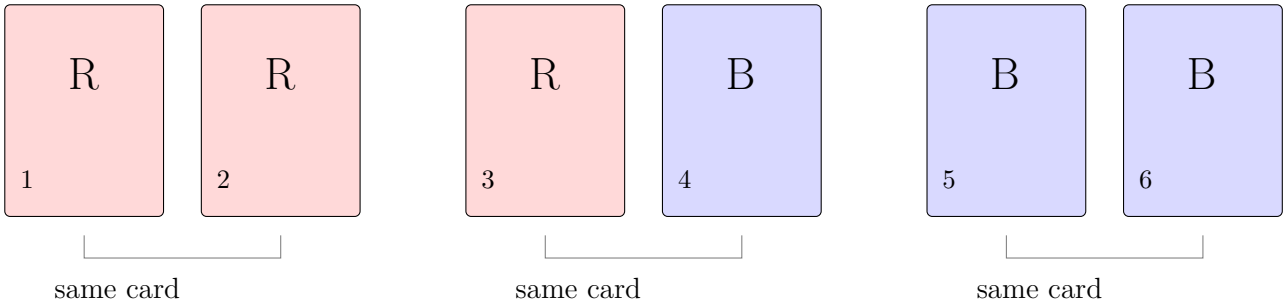
Problem 4: Compare this chooser with drawing from all six face tickets and keeping red-showing rounds. Do they give the same chances for the hidden color?

two-ticket chooser

six-ticket chooser

Return to all six face tickets. Lay each card's two tickets on it, so removing a card takes both.

Problem 5: Remove exactly one card. Can you make red showing hide red for certain? Hide blue for certain? Give the two hidden colors the same chance?



Problem 6: Keep the cards and choose two tickets for the cup. Red must sometimes show. Find every pair that makes a red clue give each hidden color the same chance.